

MLFQ Scheduling

- Multilevel Feedback Queue (MLFQ) scheduling simulation
- Project 2

Idea

- Multiple queues with different priorities
- Higher queue → higher priority

Queues Used

- Q0 $\rightarrow$ Time slice = 1
- Q1 $\rightarrow$ Time slice = 2
- Q2 $\rightarrow$ Time slice = 4

Working

- Process starts in Q0
- If it uses full time → moves down
- Scheduler picks highest priority process

Implementation

- Array-based processes
- Each process stores AT, BT, RT, queue level

Metrics

- Completion Time (CT)
- Turnaround Time (TAT)
- Waiting Time (WT)
- Average TAT and WT

Visualization

- Timeline format:
- $T0:P1(Q0) \rightarrow T1:P2(Q0)$
- Shows execution and queue levels

SCREENSHOT:

==== MLFQ Scheduling ====

Timeline:

| T0:P1(Q0) | T1:P2(Q0) | T2:P3(Q0) | T3:P1(Q1) | T4:P1(Q1) | T5:P2(Q1) | T6:P2(Q1) | T7:P3(Q1) | T8:P3(Q1) | T9:P1(Q2) | T10:P1(Q2) | T11:P3(Q2) |

PID	AT	BT	CT	TAT	WT
P1	0	5	11	11	6
P2	1	3	7	6	3
P3	2	4	12	10	6

Average Turnaround Time = 9.00

Average Waiting Time = 5.00

Conclusion

- Better for short processes
- Dynamic priority adjustment
- Simple and effective scheduling